

STEPS TO INSTALL OPENROAD TOOLS

1. If u have windows machine, download virtualbox box from <https://www.virtualbox.org/wiki/Downloads>
2. In virtual box, create a new machine.
 - Choose desired name and location
 - Type-Linux
 - Version-Redhat (32/64 bit)
 - Based on your computer RAM, assign half of the RAM to the virtual machine. At least 4 GB must be allotted.
 - Keep everything default and go forward.
 - Allot dynamic memory based on your computer memory. At least 40 GB is preferred.
3. Download CentOS 7 iso from <https://www.centos.org/download/>
4. In your newly created virtual machine, in the storage section, click on empty optical drive and insert the downloaded ISO. Finally, start the virtual machine.
5. Installing CentOS:
 - Select your language preference, Region, and keyboard language.
 - In software selection, select server with GUI.
 - In System, for installation destination, you can select automatic partition or manual partition.
 - Click on begin install
 - Setup your password and create user.
6. Installing OpenROAD from terminal:
 - Cloning repository:
 - git clone --recursive <https://github.com/The-OpenROAD-Project/OpenROAD-flow.git>
 - cd OpenROAD-flow
 - Installing docker:
 - Install needed packages:
 - sudo yum install -y yum-utils device-mapper-persistent-data lvm2
 - Configure the docker-ce repo:
 - Sudo yum-config-manager --add-repo <https://download.docker.com/linux/centos/docker-ce.repo>
 - Install docker-ce:
 - sudo yum install docker-ce
 - Add your user to the docker group with the following command.
 - sudo usermod -aG docker \$(whoami)
 - Set Docker to start automatically at boot time:
 - sudo systemctl enable docker.service
 - Finally, start the Docker service:
 - sudo systemctl start docker.service
 - Building docker:
 - ./build_openroad.sh
7. Installing KLayout:
 - yum group install -y "Development Tools" && \
 - yum update -y && yum install -y libffi-devel python3 tcl-devel which time && \
 - yum localinstall https://www.klayout.org/downloads/CentOS_7/klayout-0.26.4-0.x86_64.rpm -y
8. Running sample RTL2GDS flow:
 - Start an interactive shell in a docker container using your user credentials
 - docker run -it -u \$(id -u \${USER}):\$(id -g \${USER}) openroad/flow bash
 - Update your container environment using your environment file in my case it is env.sh
 - source ./env.sh
 - sample design flow:
 - cd flow (in docker container)
 - uncomment the desired design from Makefile
 - make DESIGN_CONFIG=./designs/path.design/name.mk (in docker container)
 - Copying gds file from docker container to host (in working area):
 - sudo docker ps
 - docker cp <container_name>:/file/path/within/container /file/path/in/host
 - klayout <file.gds>